

6.PROJECT TESTING PHASE

The Testing Phase ensures that FitBuddy works correctly and that the different components communicate properly.

Testing is performed after individual modules have been developed and integrated.

6,1 Unit Testing

Individual functions are tested independently.

Examples include:

- User input validation.
- Database operations.
- Workout generation.
- Nutrition tip generation.
- Feedback processing.

6.2 Integration Testing

Integration testing verifies communication between different components.

For example:

Frontend

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FastAPI

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Validation

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Gemini

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Generated Plan

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SQLite

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Result Page

The complete flow is tested to ensure that information is transferred correctly.

6.3 API Testing

FastAPI automatically provides Swagger documentation at:

<http://127.0.0.1:8000/docs>

The following APIs can be tested:

GET /api/health

POST /api/generate-workout

POST /api/submit-feedback

Different valid and invalid input combinations are tested.

6.4 User Interface Testing

The web interface is tested to verify:

- Form submission.
- Input validation.
- Workout display.
- Nutrition tip display.
- Feedback submission.
- Updated plan display.
- Responsive layout.

6.5 Database Testing

Database testing verifies that:

- Users are correctly stored.
- Workout plans are stored.
- Feedback is stored.
- Updated plans are stored.
- Users can be retrieved.
- Administrative operations work correctly.

6.6 Authentication Testing

The administrator page is tested with:

- Correct username and password.
- Incorrect username.
- Incorrect password.
- Missing credentials.

Unauthorized users should not be allowed to access the administrator dashboard.

6.7 AI Testing

AI-generated output is checked for:

- Seven-day structure.
- Relevant exercises.
- Correct user goal consideration.
- Appropriate workout intensity.
- Recovery guidance.
- Safety information.

The project also includes a mock-AI mode so the application can be tested without requiring a Gemini API key.